

Graded window coarsening: a per-row work bound for the operator fit that needs no estimate of the kernel’s width

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Abstract

A row of the LG-PSF operator is fitted on its *window*, every column dof inside τ_w times the caller’s a-priori ellipsoid, and the fit’s cost is linear in the window’s point count. The window is only as good as the prior: on the maintainer’s ice-sheet Hessian at continental scale a few hundred rows whose prior was too wide by decades had windows of about 90,000 points, sat on one rank, and that rank’s fit took 5,900 s against a mean of 75 s. This note records the fix. The fit’s design matrix is a mass-weighted quadrature of the model’s action on the probes, so subsampling the window is choosing a coarser quadrature, and one quadrature – cells graded by distance from the row’s centre, size at most ε times that distance, floored at single points – is adequate for every kernel width at once. Each cell becomes its mass-weighted centroid, its summed mass and the mass-weighted mean of every probe field: the in-cell probe sum is kept exactly, only the basis function’s variation over the cell is approximated. The deployed support is the full window; only the fit’s quadrature on it is coarsened, with controlled error, and the two finalists are re-scored on the full window before the baseline guard decides. Measured on sixty real rows with windows of 250 to 100,000 points, $\varepsilon = 0.1$ keeps every guard decision and 97% of ladder levels for 11–54 $\times$ less wall time on the largest windows. Whitening is as in `varpro-whitening-notes.pdf`; the measurements are `experiments/window-coarsening.md`.

1 The fit is a quadrature

Fix a row ρ with mass m_ρ and its window of K column points x_j with masses $m_j = (M_2)_{jj}$. Probes z_q , $q = 1, \dots, k$, are restricted to the window and the data are $y_q = (Hz_q)(\rho)$. (The whitening note indexes probes by ℓ ; here ℓ is reserved for the angular order of a mode.) After noise whitening the row fit is the ordinary least-squares problem

$$\hat{y}_q \approx \sum_i c_i A_{qi}(\theta) + s B_q, \quad A_{qi}(\theta) = \sqrt{m_\rho} \sum_{j=1}^K m_j z_{jq} \phi_i(x_j; \theta), \quad B_q = \sqrt{m_\rho} z_{\text{spike},q}, \quad (1)$$

with $\hat{y}_q = y_q / \sqrt{m_\rho}$, $\phi_i(\cdot; \theta)$ the Laguerre–Gaussian modes in the ellipsoid parametrized by θ , and s the spike coefficient. The unknowns (θ, c, s) are fitted from k equations indexed by the probes. The window enters only through the sum over j in A_{qi} , a mass-weighted quadrature of $\int z_q(x) \phi_i(x; \theta) dx$ over the window; the points are never regressed on. Two things follow. The cost of the fit is linear in K (every basis evaluation is over K points, and basis evaluation is about half of a row fit), and “subsample the window” means “choose a coarser quadrature of the model’s action on the probes”. A coarser quadrature is then judged as a quadrature: by the error in the columns A_{qi} at the θ the search visits.

2 Aggregation: what stays exact

Partition the window into cells C and replace each by one quadrature point,

$$m_C = \sum_{j \in C} m_j, \quad x_C = \frac{1}{m_C} \sum_{j \in C} m_j x_j, \quad z_{Cq} = \frac{1}{m_C} \sum_{j \in C} m_j z_{jq}, \quad (2)$$

so that the coarse column is

$$\tilde{A}_{qi}(\theta) = \sqrt{m_\rho} \sum_C m_C z_{Cq} \phi_i(x_C; \theta) = \sqrt{m_\rho} \sum_C \left(\sum_{j \in C} m_j z_{jq} \right) \phi_i(x_C; \theta). \quad (3)$$

The mass and the probe sum of every cell are preserved, $\sum_C m_C = \sum_j m_j$ and $\sum_C m_C z_{Cq} = \sum_j m_j z_{jq}$ (to 10^{-14} relative in every test). What is approximated is the basis function's variation over the cell, by the midpoint rule at the centroid. Expanding ϕ_i about x_C , one cell's error is

$$\sum_{j \in C} m_j z_{jq} \phi_i(x_j) - \left(\sum_{j \in C} m_j z_{jq} \right) \phi_i(x_C) = \nabla \phi_i(x_C) \cdot d_{Cq} + O(h_C^2), \quad d_{Cq} = \sum_{j \in C} m_j z_{jq} (x_j - x_C), \quad (4)$$

with h_C the cell's size and d_{Cq} the probe's *dipole* inside the cell. The centroid makes the in-cell first moment of the masses vanish, $\sum_{j \in C} m_j (x_j - x_C) = 0$; so for a probe that is smooth over the cell, $z_{jq} = z_{Cq} + O(h_C)$, the dipole is $O(h_C^2)$ and the error is second order. For a white-noise probe, the default, the dipole is as large as the monopole and the error is *first* order in h_C – but zero-mean and independent across probes, so it enters the regression like noise added to the design rather than like a bias. Measured, as the relative error of the worst design column per ladder level at a fixed θ over $\varepsilon = 0.05$ to 0.3 : slopes of 1.0 in ε for white-noise probes (0.84 – 1.12 on a 3×10^5 -point disc, 0.97 – 1.21 on real windows) and 2.0 for smooth probes (1.7 – 2.2). At $\varepsilon = 0.1$ the white-noise column error runs from 1.5% at level 0 to 10% at level 5 , and the cross-validation score at a fixed θ moves by about 8×10^{-3} on the largest windows. A dipole-corrected cell – carry d_C and add $\nabla \phi_i(x_C) \cdot d_C$ to the design – would restore second order; the numbers do not ask for it.

3 The graded cell rule

The rule. Work in the whitened coordinates of the window frame, $u = L_w^{-1}(x - \mu_w)$ (§4). Build a 2^N -tree on the bounding box of $\{u_j\}$, splitting at the midpoint of every axis; a node is split while it holds more than one point and

$$\text{diag}(\text{box}) > \varepsilon d(\text{box}), \quad d(\text{box}) = \text{dist}(0, \text{box}), \quad (5)$$

the distance from the origin to the box's nearest point, zero when the box contains the origin. Every leaf is a cell. So a multi-point cell at distance r from the centre has size $h \leq \varepsilon r$, and cells shrink to single points as they approach the centre.

Uniform in the width. For a Gaussian $g(r) = e^{-r^2/2\sigma^2}$ the midpoint rule on a cell of size h at distance r errs by about $h |g'(r)|$,

$$h \frac{r}{\sigma^2} e^{-r^2/2\sigma^2} = \frac{h}{r} t e^{-t/2} \Big|_{t=r^2/\sigma^2} \leq \frac{2}{e} \frac{h}{r} = 0.74 \frac{h}{r}, \quad (6)$$

the maximum over σ at fixed r falling at $r = \sqrt{2}\sigma$, where the kernel is still e^{-1} of its peak. With $h \leq \varepsilon r$ the error is at most 0.74ε of the peak for every width σ simultaneously. The rule needs no estimate of the kernel's width – which is the point, since the row whose window is far too wide is exactly the row whose prior width is wrong.

The modes' smoothness. A Laguerre–Gaussian mode of radial degree p and angular order ℓ is a polynomial of degree $2p + \ell$ times the Gaussian: its finest structure is at scale $\sigma/(p + \ell)$, it lives near $r \sim \sigma$, and beyond that the Gaussian envelope takes over. Three regions:

- *The singleton near field*, $r \lesssim h/\varepsilon$ on a mesh of spacing h . A box containing the centre has $d = 0$ and splits down to single points, and no two points merge closer than h/ε , since a two-point cell has diagonal at least h . The fit sees every point.
- *The mode's own radius*, $r \sim \sigma$: cells of size $\varepsilon\sigma$ against structure at scale $\sigma/(p + \ell)$. The midpoint rule on a smooth integrand errs by the square of the ratio, $(\varepsilon(p + \ell))^2$; resolving the structure at all asks for

$$\varepsilon(p + \ell)_{\max} \lesssim 0.5. \quad (7)$$

A shell ladder to level L holds every mode with $2p + \ell \leq L$, so $(p + \ell)_{\max} = L$: at the ladder's top level of 5, $\varepsilon \leq 0.1$.

- *Beyond*, $r \gg \sigma$: cells grow as εr but the mode decays as a polynomial times $e^{-r^2/2\sigma^2}$, and (6) already covers it.

A coarser ε under-resolves the high modes; their aliased columns look like noise to the cross-validation, so the ladder stops lower. That is the conservative failure, and every measured casualty at $\varepsilon \geq 0.15$ is of this kind – a ladder that stopped early, never a fit that wandered.

The count. In 2D the dyadic annulus $[r, 2r]$ has area $3\pi r^2$ and cells of size εr , so about $3\pi/\varepsilon^2$ cells per annulus and

$$K_c \sim \frac{3\pi}{\varepsilon^2} \log_2 \frac{R}{h} \quad (8)$$

for a window of radius R on a mesh of spacing h : logarithmic in the window's size, which is what makes the fit's cost scale under mesh refinement, and finite in 3D, where an octree gives about $4\pi/\varepsilon^3$ per shell. Two corrections matter at practical sizes. The near field is a *singleton core* of about $7\pi/\varepsilon^2$ points – nothing merges within h/ε of the centre, and the dyadic quantization keeps most cells single out to a few times that – so the law needs a floor: at $\varepsilon = 0.05$ the core is nearly the whole count below 10^4 points, and no window below about 2,000 points coarsens at all. Where the grading bites, the constant sits above 3π , because the rule bounds the box *diagonal* and dyadic halving leaves leaves between half and the full allowed size. Measured on ball windows of 10^5 grid points, K_c over (8) runs from 0.76 at $\varepsilon = 0.05$ (floor still binding) to 1.7 at $\varepsilon = 0.3$; a 10^5 -point window becomes 21,600 cells at $\varepsilon = 0.05$, 7,700 at 0.1 and 4,000 at 0.15.

4 Why the window frame

The window is the ellipsoid $\{x : (x - \mu_w)^\top \Sigma_w^{-1} (x - \mu_w) \leq 1\}$ with $\Sigma_w = \tau_w^2 \text{cap}_a(\Sigma_{\text{prior}})$, where cap_a floors the minor axes so that the axis ratio is at most a (`window_aspect_cap`). It is already

the object that encodes how much of the prior’s shape the library trusts, so the cells are built in its whitened coordinates: with $\Sigma_w = L_w L_w^\top$,

$$u = L_w^{-1}(x - \mu_w), \quad (9)$$

and (5) grades by $|u|$, the Mahalanobis distance of Σ_w . A ball window, $a = 1$, makes L_w a multiple of the identity and the grading Euclidean. A user who lifts the cap and trusts an anisotropic prior gets the grading in that prior’s own metric for free: a kernel aligned with the window is resolved isotropically in the window’s coordinates, and a kernel rotated against the window loses resolution along its major axis by the window’s aspect ratio – the loss the fit already accepts through the cap. One tree, no new notion of trust. The cells are axis-aligned boxes in the Cholesky coordinates (parallelograms in physical space when L_w has off-diagonal entries); the grading is the metric of Σ_w regardless.

5 Protected positions

Three kinds of position stay singleton cells, copied bit for bit: a node holding one is split until that point is alone.

The spike. Its column in (1) is $B_q = \sqrt{m_\rho} z_{\text{spike},q}$, with no mass in it: the whitened extra is $\hat{e} = \sqrt{m_\rho} M_2^{-1/2} e$ and e is one-hot, so the $M_2^{-1/2}$ cancels against the $M_2^{1/2}$ in $\hat{z}$. A singleton reproduces the column exactly and keeps the meaning of s , of the spike’s measure and of the assembled diagonal entry; merged into a cell, the spike’s row would carry the cell’s mass and s would be silently rescaled.

The support of extra bases. For a general extra column E the design column is the plain sum $\sqrt{m_\rho} \sum_j z_{jq} E_j$, without masses, because an extra is a functional and not a kernel evaluation. Aggregation preserves $\sum_j m_j z_j$, not $\sum_j z_j E_j$, so a general E is not coarsened consistently by (2). What makes the spike work generalizes with no special case: `coarsen.window` takes a set of *protected positions* and keeps each a singleton; `fit.operator` passes the spike, a direct caller with extras passes the union of their supports, and sparse extras are then exact. If the protected set would exceed `coarsen.above`, the row is left uncoarsened rather than refused, and `fit.points` says so.

The farthest point. $R_w = \max_j |x_j - \mu_0|$ (`window.radius`) is both the window-containment admissibility bound on a released centre and the ladder’s top rung. A centroid lies strictly inside its cell’s hull, so without protection R_w would silently tighten; the point farthest from the centre in physical distance stays a singleton (ties to the smallest window position). The bottom rung, `local.spacing`, is preserved by the singleton near field of §3. The spike and the farthest point are distinct unless the window has one point, so the coarse set always has at least two.

6 Deployment, and the full-window re-score

The deployed row is the fitted kernel’s τ -ellipsoid intersected with the full window,

$$\text{support} = \text{window} \cap \{ \|u_\theta\| \leq \tau \}, \quad (10)$$

and nothing in the coarsening writes the window. So the guarantee is: *the deployed support is the fit window; the fit’s quadrature on it is coarsened with controlled error*. Near the centre the two quadratures agree exactly, far out the kernel is small, and every deployed point lies in a cell the fit saw – the out-of-window extrapolation pathology that forced “deployed support = window” cannot recur inside it.

What can go wrong is a score. On a coarsened row the cross-validation score that drives the ladder and the baseline guard is a coarse-quadrature score, and it is optimistic: the coarse quadrature adds noise to the design, and the coarse baseline moves too. After the search, the two finalists – the winner and the baseline – are re-scored on the full x, m, z (one `linear_cv_score` each, $O(Km)$ for m modes, negligible against a fit; the coefficients are not refit). `score` and `baseline_score` report those numbers, and the guard “ship iff score < baseline” is taken on them. Measured on the fifteen largest windows: a model’s coarse score differs from its full-window score by up to 0.018 at $\varepsilon = 0.1$ and 0.056 at 0.3; the guard taken on coarse scores flips no decision at $\varepsilon \leq 0.15$ and claims 2 false wins over the baseline at 0.2 and 4 at 0.3, never the reverse. With the re-score the quadrature’s optimism can cost a row its fit, never ship a bad one.

Released centres. The remaining gap is a candidate whose centre is fitted and displaced by $\|\mu - \mu_0\|$ from the grading’s centre: it sits on cells of size $\varepsilon \|\mu - \mu_0\|$, and a needle narrower than those cells is under-resolved there – its coarse columns are aliased, its coefficients can inflate, and its coarse score says nothing about the fine evaluation at deployment. Coarsened rows therefore arm the resolution rule (`resolution_eps`, set to `coarsen_eps`): a released candidate is admissible only if

$$\min \text{axis} \geq \varepsilon \|\mu - \mu_0\| \max(1, (p + \ell)_{\max}), \quad (11)$$

the floor of 1 being the pure Gaussian, whose structure is at σ itself. Pinned candidates are untouched, and the re-score is the consistency check behind the rule: an aliasing artifact scores well on the cells and badly on the points. Dead rows are safe by themselves – zero data give zero coefficients whatever the design.

7 Determinism

`coarsen.window` is a pure function of its arguments: no threads, no randomness, no dependence on allocation. The split is canonical (midpoints; a point on a split plane goes to the upper child; empty children are dropped), cells are emitted sorted by the smallest window position they contain, and each cell accumulates (2) in ascending position. The same inputs give bit-identical outputs, so the library’s bit-identity across thread counts and runs is untouched. In the distributed fit each rank’s window array is merged in ascending global id and is therefore independent of the rank layout; a pure function of that array in that order is rank-independent too, and the MPI gate with coarsening on passes bitwise at 1, 2 and 4 ranks (564 of 600 rows coarsened, 20,305 window points fitted on 13,046 cells). Setting `coarsen_above = 0` restores the pre-coarsening path byte-identically.

8 Using it

Two knobs on `OperatorFitConfig` and one diagnostic:

- `coarsen_above` (default 3,000; 0 switches it off): windows with more points than this are coarsened. A few thousand is the useful range – below about 2,000 points the grading is the identity and buys nothing. On by default since 2026-09-08, after field validation on a continental ice-sheet Hessian took the worst rank’s fit from 2,015 s to 588 s at the same prior, with the operator, the QC ladder and the deployed solve unchanged. Fits on rows above the trigger are therefore not bitwise identical to releases before that date.
- `coarsen_eps` (default 0.1): the grading ratio of (5); (7) at the ladder’s top level of 5.
- `FitDiagnostics::fit_points`: per row, what the fit ran on – the window size, or the coarse count when the trigger fired. Whether the bound is binding is readable here and nowhere else (`DistFitResult::fit_points_total` sums it).

The ε sweep of `experiments/window-coarsening.md` is Table 1: sixty real rows of the heat-equation inverse problem at four mesh spacings, windows of 246 to 99,856 points, 60 probes, a shell ladder to level 5 decided by cross-validation alone. Every coarse fit is re-scored on the full window. “Fits” is the fraction of rows whose fit beats the full-window baseline, against 0.87 for the full fits; “same level” the fraction shipping the full fit’s ladder level; “max Δ score” the worst full-window score change over the sixty rows; “max coarse–full” the largest gap between a model’s coarse-quadrature score and its full-window score on the fifteen largest windows, and “false wins” how often the guard taken on coarse scores would have shipped a model the full-window scores reject.

ε	median K_c/K	fits	same level	max Δ score	max coarse–full	false wins	wall ratio
0.05	0.864	0.87	0.98	+0.0029	0.0072	0	0.235
0.1	0.493	0.87	0.97	+0.0108	0.0181	0	0.110
0.15	0.305	0.85	0.88	+0.1366	0.0256	0	0.072
0.2	0.219	0.78	0.93	+0.0344	0.0381	2	0.054
0.3	0.129	0.72	0.75	+0.0730	0.0556	4	0.036

Table 1: The ε sweep on sixty real rows, windows of 246 to 99,856 points. Full fits: 0.87 of rows beat the baseline.

$\varepsilon = 0.1$ keeps every guard decision and the ladder level on 97% of rows, the fitted axes within 10^{-4} in $\log_{10}$ and the orientation within 0.25° , c_0 within 0.02%, for $9\times$ less wall time overall and $11\text{--}54\times$ on the 10^5 -point windows (23–30 s to 0.5–1.4 s). $\varepsilon = 0.15$ loses the first fit to an early ladder stop; $\varepsilon \geq 0.2$ loses 10–17% of the fits and makes the coarse score dishonest. The motivating case, a prior too wide by decades, is the easy one: the kernel then lies inside the singleton core and is fitted exactly (on the rows with a $4\times$ widened prior the score is within 0.0003 of the full fit’s at every ε up to 0.2). The rows that feel ε are honest-prior rows on fine meshes, whose kernels fill their windows.

Cost: a cell costs what a point costs (5.7×10^{-6} against 6.0×10^{-6} s per candidate), so the saving is exactly the cell ratio K_c/K ; by (8) a 10^5 -point row at $\varepsilon = 0.1$ costs less than a 10^4 -point row does today. The coarsening itself is $O(K \log K)$ and is not the bottleneck.